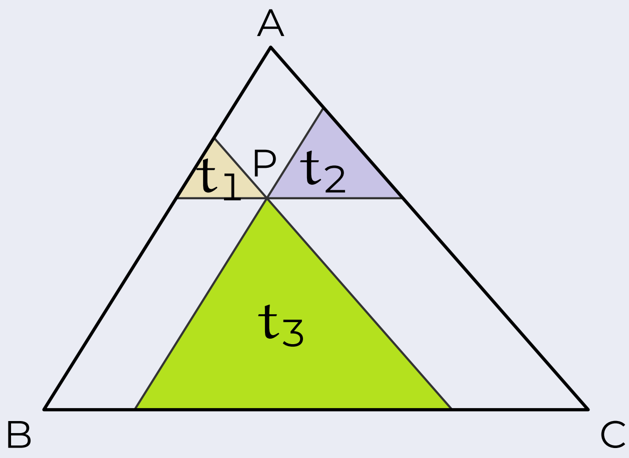
INDIAN INSTITUTE OF INFORMATION TECHNOLOGY RAICHUR
MATHEMATICS CLUB
PROBLEM NUMBER: WP2601
DATE: JANUARY 1, 2026
DUE DATE: JANUARY 16, 2026

WEEKLY PROBLEM

EXERCISE.

Part A:

A point P is chosen in the interior of such that when lines are draw through P parallel to the side of $\triangle ABC$, the resulting smaller triangles t_1,t_2 andt_3 , as shown in the figure, have area 4, 9 and 49 respectively. Find the area of $\triangle ABC$.



Part B:

Extend your analysis when the area of triangles t_1,t_2 and t_3 formed by these lines have area A_1,A_2 and A_3.

Part C:

What should be the ratio of the areas A_1,A_2 and A_3 such that the point P coincides with the centroid of the $\triangle ABC$.

(Scan this to submit solution!)

Top Submissions for last Week's Problem (WP2505):

1. **Hamza Nazim Siddiqui**
   AD24B1023

Wishing you all Happy New Year !!